The Anatomy of a Decentralized Prediction Market: Microstructure Evidence from the Polymarket Order Book

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May 14, 2026

JEL Classification: G14 (Information and Market Efficiency; Event Studies; Insider Trading), G12 (Asset Pricing; Trading Volume; Bond Interest Rates), G19 (General Financial Markets), C58 (Financial Econometrics), L86 (Information and Internet Services; Computer Software).

Keywords: Prediction markets, microstructure, limit order book, Polymarket, decentralized finance, on-chain data, trade-direction inference, effective spread, Kyle's λ, wash trading.

Abstract

We study the microstructure of Polymarket, the largest on-chain prediction market, using a continuous tick-level archive of the public WebSocket order-book feed (30 billion events over 52 days) joined to the authoritative on-chain trade record. On a pre-registered stratified panel of 600 markets we report eight stylized facts: a longshot spread premium; a depth-concentration profile closer to a uniform geometric grid than to the top-of-book pattern often assumed for prediction markets; a null block-clock alignment effect; broad maker-wallet diversity with a concentrated tail; category-conditional differences in effective spread; a sub-50 ms median archive-ingestion delay with a multi-second tail; a self-counterparty wash share with median 1% and a 22% upper tail, where the comparison to the network-classifier benchmarks of Cong et al. [2023] on unregulated cryptocurrency token exchanges is a sanity bound rather than an apples-to-apples reference because the venues face different wash incentives; and a cross-sectional depth profile explained by market duration, price level, and volume, with no residual time-to-close effect once those three are controlled for. The paper also contributes a measurement result: trade direction inferred from Polymarket's public order-book feed agrees with on-chain ground truth only $\sim$ 59% of the time (volume-weighted across the top-100 stratum and four disjoint 7-day windows; panel mean 0.615, market-clustered bootstrap 95% CI $[0.58, 0.65]$, IQR $[0.53, 0.68]$), barely above the 50% chance baseline that direction-dependent microstructure measures need. On the comparable subset of the top-100 panel, the effective half-spread changes sign between feed- and on-chain trade directions on 67% of markets in the first 7-day window (50% in a second non-overlapping window), and Kyle's λ changes sign on 60% (43% in the second window); both windows fail to recover the on-chain sign at anything close to the $\sim$ 80% rate that Lee-Ready achieves on equity venues. Microstructure work on Polymarket therefore needs to source trade direction from on-chain `OrderFilled` events; we release a replication package that performs the join.

1 Introduction

Prediction markets aggregate dispersed beliefs into a single price that, in equilibrium, behaves like a probability. The empirical literature on these markets has historically focused on price-level questions: forecast accuracy, calibration against realised outcomes, the longshot bias, and the extent to which informed and uninformed traders coexist on the same venue [Wolfers and Zitzewitz, 2004, Manski, 2006, Page and Clemen, 2013]. Yet microstructure is what determines the trading cost of holding an informational position, and the wider the cost, the smaller the informational signal that survives to the price. A prediction market with noisy microstructure produces noisier prices than the headline-level aggregation literature implicitly assumes. The microstructure of prediction markets, however, has remained under-studied: limit-order-book depth profiles, spread decompositions, and the behaviour of liquidity providers near resolution were largely unobservable on the early venues, which used scoring rules (IEM, the logarithmic market scoring rule of Hanson 2007) or sparse parimutuel pools. Two recent Polymarket-specific contributions sit close to the present paper but ask different questions. Rahman et al. [2025] survey decentralised prediction-market microstructure as a methodological domain, framing the open questions across venues without bringing matching tick-level data. Tsang and Yang [2026] run a single-event time-series microstructure study on the 2024 US presidential election market, documenting episode-level patterns around Biden's withdrawal, the September debate, and the October whale trades, and reporting a roughly order-of-magnitude decline in Kyle's λ as the market matured; the unit of analysis is one market over many time slices, with λ estimated on aggregated rather than tick-level data. The present paper is the cross-sectional, tick-level complement: 600 markets observed simultaneously over a single 28-day scrape window, with microstructure measures computed on the full event tape and a direct on-chain trade record. To our knowledge, neither authoritative on-chain trade direction nor a continuous off-chain order-book archive at tick resolution has been brought to bear at the cross-sectional scale we use here. The classical microstructure references behind our measurement choices are O'Hara [1995] and Hasbrouck [2007] for order-book theory, Foucault et al. [2013] for the liquidity-provider angle, and Huang and Stoll [1997], Madhavan et al. [1997] for spread decomposition.

Polymarket changed this. Since 2021 the venue has run a limit-order-book exchange on Polygon, settling in USDC against an on-chain conditional-token contract. Our data covers 52 calendar days of the public order-book feed (2026-02-21 to 2026-04-15) at tick resolution: 30,287,264,368 events across 385,198 distinct markets. A 28-day overlap window (2026-02-28 to 2026-03-27) is mirrored on chain, where the CTF Exchange V1 contract logs 255 million `OrderFilled` events whose payloads identify both counterparties and the aggressor side. We use the off-chain feed for quote dynamics and the on-chain record for trade direction, and study both on a pre-registered, stratified 600-market panel.

The paper has two empirical contributions, ordered by weight for the literature.

Trade-direction inference from Polymarket's public order-book feed is noise-dominated. Six standard direction-dependent measures (effective spread, realized spread, Kyle's λ, Amihud, Roll, Abdi-Ranaldo) need an aggressor sign for each trade, and empirical practice on equity venues sources that sign from a quote-driven feed via Lee-Ready or its variants [Lee and Ready, 1991, Ellis et al., 2000]. The Polymarket public order-book feed does not, by itself, expose enough information to do this reliably: the `change_side` field marks which side of the book moved, not which side initiated the trade. Sign agreement between feed-inferred and on-chain trade directions sits at $\sim$ 59% volume-weighted across the top-100 stratum and four disjoint 7-day windows, just above the 50% chance baseline. On the comparable subset of the top-100 panel, the effective half-spread changes sign on 67% of markets in a first 7-day window when feed-inferred direction is swapped for the on-chain record, and Kyle's λ changes sign on 60%; in a second non-overlapping window the rates fall to 50% and 43% respectively. Both windows sit in or below the chance band, well short of the $\sim$ 80% Lee-Ready accuracy documented on Nasdaq. Any Polymarket microstructure result that depends on trade direction therefore needs to source it from on-chain `OrderFilled` events. We release a replication package that performs the join.

Eight cross-sectional stylized facts on the panel. Quoted half-spread shows the longshot premium familiar from the racetrack literature; the L2 depth profile is closer to a uniform geometric grid than to the top-of-book pattern usually assumed for prediction markets; quote-update timing does not cluster on Polygon block boundaries to a meaningful degree; maker-wallet concentration is diverse on most markets with a thin concentrated tail; effective spreads differ by category; archive-ingestion latency is tightly distributed around 41.5 ms with a multi-second p_{99} tail; the self-counterparty wash share has a 1% median and a 22% maximum across the panel; and cross-sectional depth is explained by market duration, price level, and volume, with no residual time-to-resolution effect once those three are conditioned on. None of these eight facts requires the on-chain join, and each is reported on the full 600-market pre-registered panel.

The two contributions are complementary. The eight stylized facts characterise Polymarket's microstructure on its own terms. The measurement result sets the conditions under which any trade-direction-dependent statement about Polymarket can be trusted.

Section 7 carries the most counter-intuitive finding; readers more interested in Polymarket as an empirical object will care most about SF1 (longshot premium), SF2 (depth concentration), and SF8 (cross-sectional depth structure).

2 Institutional Background

2.1 Conditional tokens and binary markets

A Polymarket binary market resolves to one of two outcomes. The exchange records the outcomes as ERC-1155 conditional tokens issued by Gnosis's Conditional Tokens Framework (CTF) on Polygon. Each market has a YES token and a NO token; the contract guarantees that one YES + one NO can always be redeemed for one USDC after resolution. As a consequence, in equilibrium the YES price plus the NO price equals one and either side can be priced as the probability of the corresponding outcome.

Liquidity exists for both sides. A trader who is long YES and believes the implied probability is too high can either sell YES on the YES book or buy NO on the NO book. The two strategies differ in inventory effects and slippage but are economically equivalent at mid. Throughout this paper we report measures on the YES side; YES and NO books are structural mirrors of each other and the panel artifacts contain both.

2.2 CTF Exchange contract architecture

Trading happens through the Polymarket CTF Exchange smart contract, which matches signed off-chain orders against resting liquidity and executes settlement on chain. The contract has two versions. V1 (`0x4bFb41d5B3570DeFd03C39a9A4D8dE6Bd8B8982E`) ran from launch through the end of April 2026 and is the version our 28-day scrape window targets. V2 (`0xE111180000d2663C0091e4f400237545B87B996B`) launched in early April 2026 with a hard cutover; V1 orders were rejected after the switch. Both versions emit the same `OrderFilled` event signature, with topic hash `0xd0a08e8c493f9c94f29311604c9de1b4e8c8d4c06bd0c789af57f2d65bfec0f6`.

The event payload carries (`orderHash`, `maker`, `taker`, `makerAssetId`, `takerAssetId`, `makerAmountFilled`, `takerAmountFilled`, `fee`). The asset ids identify which side held USDC: `makerAssetId == 0` means the maker posted USDC and the taker is the seller of the conditional token; `takerAssetId == 0` means the taker posted USDC and is therefore the buyer. This binary asset-id field is the on-chain ground truth for trade direction that we use throughout the paper.

2.3 Polygon block timing and settlement

Polygon mainnet produces a block roughly every two seconds. Trade settlement is final to within Heimdall's 256-block reorg buffer, so we scrape only blocks at depth ≥ 256 to avoid reorg contamination. Block timestamps are monotonic but not perfectly spaced; for our window the empirical mean inter-block interval is 2.0 s with a small variance, well below the 60-second sample step used in the panel compute. Where we need to attach a wall-clock timestamp to an on-chain trade we use linear interpolation from a single anchor block, which is accurate to within a few seconds over the 28-day window.

2.4 Public WebSocket feed structure

In parallel with on-chain settlement, Polymarket broadcasts a public WebSocket feed that exposes order-book state to any subscriber. The feed emits two event types. `book_snapshot` carries a complete L2 snapshot of one side of one market and is sent on subscription and at irregular intervals thereafter (0.80% of events in our archive). `price_change` carries a delta against the last known book state: a single triple `(change_price, change_side, change_size)` that updates one level on one side. A `change_size` of zero means the level is empty; a non-zero size is the new resting quantity at that price. These events are 99.20% of the archive.

The feed does not expose taker identity. The `change_side` field labels which side of the book the level update applies to, not which side initiated the trade that produced the update. This distinction drives the measurement gap of Section 7: an inference algorithm working from the feed alone cannot reliably reconstruct the aggressor sign because the underlying field never encodes it.

2.5 Aggressor-sign dependence of standard measures

Most direction-dependent microstructure measures take the form

$$M = \mathbb{E}[\mathrm{sign}_t \cdot f(\mathrm{price}_t, \mathrm{mid}_t, \mathrm{size}_t)]\,,$$

where $\mathrm{sign}_t \in \{-1, +1\}$ is the aggressor sign for trade t and f is the per-trade contribution. Effective half-spread ($f = \mathrm{price}_t - \mathrm{mid}_t$), realized half-spread, Kyle's λ regression coefficient, and the adverse-selection component of Glosten-Harris all instantiate this form. A noisy sign_t does not just attenuate M; depending on the distribution of f across trades, it can flip the sign of M entirely. The 59% sign-agreement rate of Section 7.1 is what produces the 67% effective-spread and 60% Kyle's λ sign-flip rates we report on the top-100 panel.

2.6 Comparison to equity-market microstructure

Equity-market benchmarks are what give several of our findings their scale. Ellis et al. [2000] report that the Lee-Ready algorithm correctly classifies 81% of trades on Nasdaq, and the original Lee and Ready [1991] study reports a similar accuracy on NYSE TAQ; the $\sim$ 59% feed-driven inference rate on Polymarket is therefore about 22 percentage points below the equity benchmark on the same algorithm. Effective half-spreads on liquid US equities post-decimalization sit in the single-digit-basis-point range [Hasbrouck, 2007, Foucault et al., 2013]; on Polymarket, expressed in basis points relative to mid, the median quoted half-spread on the central price decile is around 200 bps, an order of magnitude wider, consistent with longer-horizon prediction-market positions and substantially smaller market-maker capital. Liquidity-provider activity on US equities is heavily concentrated in a handful of high-frequency firms that account for roughly half of trading volume [Brogaard et al., 2014]; the median Polymarket market in our panel has $\sim$ 32 effective makers (SF4), suggesting a flatter maker landscape on the venue. Wash trading on regulated equity exchanges is essentially zero (it is prohibited under SEC Rule 10b-5); on unregulated cryptocurrency token exchanges Cong et al. [2023] document wash-trade shares of 25–70% via a network-classifier approach. Polymarket sits at a median 1% under our direct-detection lower bound with a 22% upper tail. The token-exchange benchmark is a different incentive environment (volume-tied listing fees, aggregator-ranking optimisation, market-making rebates), so we treat 25–70% as a sanity bound rather than an apples-to-apples reference. Finally, collector-side ingestion latency for direct equity feeds is sub-millisecond and SIP latency is in the hundreds of microseconds [Hasbrouck, 2007]; our 41 ms median for archive ingestion is two orders of magnitude slower, but that gap reflects the archive pipeline rather than the exchange.

3 Data Construction

3.1 Off-chain orderbook archive

The off-chain data come from an external tick-level archive of Polymarket's public WebSocket order-book feed, covering 2026-02-21 16:00 UTC through 2026-04-15 08:00 UTC. We restrict the archive to the two book-state events emitted on the market channel — `price_change` and `book_snapshot`; the channel's separate `last_trade_price` event is out of scope, and trade direction is sourced from the on-chain record instead. Each row contains the raw event JSON plus two timestamps: `timestamp_received` from the exchange and `timestamp_created_at` from the upstream archive. Three hourly files are missing (2026-02-24T14, 2026-02-24T15, 2026-04-04T18); the gaps are stable and treated as missing rather than malformed throughout the analysis. Total archive: 1,262 hourly Parquet files, 30,287,264,368 rows, 623.8 GB on disk.

The schema follows the WebSocket payload for the two retained event types, with the JSON `data` string parsed on demand. We avoid eager parsing because the row count makes per-event Pydantic validation a multi-hour operation; we instead apply structured parsing only to the events that survive market-id and time-window filters.

3.2 On-chain trade scrape

We scrape `OrderFilled` events from the CTF Exchange V1 contract across the 28-day calibration window (2026-02-28 to 2026-03-27). The scraper issues batched `eth_getLogs` calls against a Polygon RPC provider, with adaptive chunk sizing that respects provider-specific rate limits (`scripts/scrape_onchain_fills.py`). Each call returns at most a few hundred events; total fills across the window are 255,425,405. The scraper writes one Parquet shard per five-thousand-block slice, deduplicates on `(transactionHash, logIndex)`, and skips blocks at depth less than 256 (Polygon's Heimdall reorg buffer).

Block timestamps are attached via linear interpolation from a single anchor block at the start of the window. We verified the interpolation error against per-block RPC calls on a 50-block stratified sample drawn across the 28-day window: the maximum absolute gap is 4 seconds, the median is 2 seconds. Both are well below the 60-second sample step used in the panel compute.

The on-chain decoder (`polydata.onchain.join.aggregate_onchain_df`) drops "split fills" where neither side is USDC (the binary maker/taker-asset-id rule for the aggressor sign breaks down when both sides hold conditional tokens). Across the 255,425,405 fills in the scrape window, 0 fills satisfy this condition; the filter is non-binding on the V1 contract during our window.

3.3 Schema reconciliation across sources

The off-chain feed is keyed on `market_id` (66-character hex `conditionId`) and `token_id` (256-bit decimal, the YES or NO ERC-1155 id). The on-chain record is keyed on `makerAssetId` and `takerAssetId`, which are the same ERC-1155 ids except that one side is set to `0` to mark USDC. The bridge between the two is the `(condition_id, yes_token_id, no_token_id)` mapping, which we pull from CLOB REST (`/markets/{conditionId}`) and cache locally. CLOB REST resolves all 385,198 archive market ids; the Gamma metadata API, which is sometimes used in the literature, indexes only 34,764 markets and is not the right source for this join.

3.4 Pre-registration of the panel

The 600-market panel is the cross-sectional unit on which the empirical work runs. We commit the selection rule before computing the panel. The pre-registration document (`docs/preregistration_plan3c.md`) specifies the volume metric, the random-stratum eligibility threshold, the random seed, and the categorisation scheme used for SF5. A deterministic build script (`scripts/build_panel.py`) emits the panel parquet, whose SHA-256 is recorded back into the pre-registration document before any analysis runs.

This discipline goes beyond the empirical-microstructure norm; the cost is one document and one hash, the benefit is that a reader can verify no market was added or removed after the analysis ran.

3.5 Replication package

All code that produced the artifacts cited in this paper is in the repository at `https://github.com/philippdubach/polymarket-microstructure`, archived for replication under DOI `10.5281/zenodo.19811426` [Dubach, 2026]. The full 624-GB WebSocket archive is not redistributed in the package; this section describes the archive's structure and we provide the panel parquet artifacts (`data/panel.parquet`, `data/panel_trade_measures.parquet`, `data/panel_quote/*`) directly, together with the on-chain scrape pipeline. A reader who wants to reproduce a single number in the paper can do so against the artifacts; a reader who wants to extend the analysis to a different window will need to capture or source an equivalent tick-level archive of the Polymarket WebSocket feed.

4 Data

4.1 Orderbook archive

Our primary input is a continuous tick-level archive of Polymarket's public WebSocket order-book feed from 2026-02-21 16:00 UTC through 2026-04-15 08:00 UTC, 52 calendar days. The archive holds 1,262 hourly Parquet files (623.8 GB on disk) and 30,287,264,368 event rows. Three hourly files are missing (2026-02-24T14, 2026-02-24T15, 2026-04-04T18); we treat the gaps as missing rather than malformed.

Each row is either a `price_change` update (99.20% of events) or a `book_snapshot` (0.80%); the payload is a JSON string parsed on demand. Two timestamps are emitted per event: `timestamp_received` (exchange-side) and `timestamp_created_at` (ingest-side). The difference captures the archive's ingestion delay (Section 5.6).

The archive contains 385,198 distinct `market_id` values (roughly twice that with the YES/NO side split). The CLOB REST endpoint resolves token-pair metadata for all 385,198. The Gamma metadata API is sparser, indexing only 34,764 markets.

4.2 On-chain trade join

Orderbook events alone do not identify the trade-aggressor sign reliably: feed-inferred sign matches the on-chain record on only $\sim 59\%$ of comparable buckets, just above the 50% chance baseline (Section 7). We therefore scrape Polymarket's CTF Exchange V1 contract (`0x4bFb41d5B3570DeFd03C39a9A4D8dE6Bd8B8982E`) for `OrderFilled` events on Polygon mainnet across the 28-day calibration window 2026-02-28 – 2026-03-27, yielding 255,425,405 fills. The aggressor sign reads directly from the event fields: $+1$ when `takerAssetId` $= 0$ (taker posted USDC and is therefore the buyer), -1 when `makerAssetId` $= 0$.

4.3 Stratified panel

Our cross-sectional panel has 600 markets in two strata. The top-100 stratum ranks markets by total on-chain USDC trade volume in the scrape window (summing across both YES and NO tokens). The random-500 stratum samples uniformly without replacement from markets with at least 100 on-chain trades that resolve to a token pair via the CLOB cache. We commit the random seed (20260424) in the pre-registration document (`docs/preregistration_plan3c.md`) before computing the panel.

Per-market metadata fields `question`, `end_date_iso`, and `closed` are pulled from CLOB REST `/markets/{conditionId}` for all 600 markets. The top-stratum volume range is \$4.56M to \$96.0M USDC; the random stratum spans 100 to 24,378 trades per market. Crypto markets dominate the panel composition (348/600, 58%), followed by Sports (142/600, 24%); Table 3 gives the breakdown.

5 Stylized Facts

We document eight stylized facts on the 600-market panel (Section 4.3). Each subsection reports a per-market distribution and, where the data support one, a regression coefficient. Underlying parquet artifacts are in the replication package.

5.1 SF1 – Longshot spread premium

We bin quoted half-spread (in basis points of mid) by per-market mean mid price into ten deciles. Median spread is $\sim$ 400 bps in the central $[0.4, 0.6]$ range and climbs to 1,300–1,800 bps for markets trading below 0.10. The pattern is asymmetric: the low-probability side is wider than the high-probability side, which echoes the longshot bias documented in the racetrack and parimutuel literature [Snowberg and Wolfers, 2010, Thaler and Ziemba, 1988] and the prediction-market longshot evidence on Iowa Electronic Markets and TradeSports surveyed by Wolfers and Zitzewitz [2004]. The direction is the same; the magnitude is not. Racetrack longshot premia are typically a few percent of stake, and prediction-market longshot biases observed on IEM and TradeSports run in the same range. The 1,300–1,800 bps full quoted spread on Polymarket's lowest- probability decile (i.e., 650–900 bps half-spread) is an order of magnitude wider, which reads less like the classical risk-love or misperception story and more like a liquidity-provision constraint: low-probability binary contracts have a bounded upside and an asymmetric downside for the maker, so the inventory-risk premium on the wide side is mechanically larger than on a continuous-payoff sportsbook market. Distinguishing inventory risk from a behavioural longshot premium would require data we do not collect (maker realised inventory and time-on-book), so we report SF1 as a Polymarket-specific quoted-spread stylised fact and leave the structural decomposition to follow-on work.

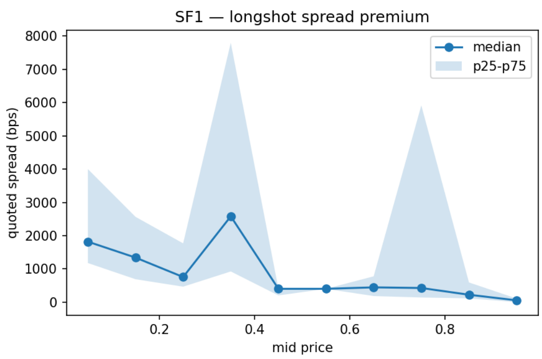

Figure 1: SF1 panel: median quoted spread (bps) per mid-price decile, 600 panel markets. Shaded band is interquartile range.

5.2 SF2 – Depth concentration

We summarize the L2 depth profile by the per-level share of cumulative top-10 depth, computed for each market across the window. The share at level k is the average per-level depth divided by average cumulative top-10 depth, where per-level depth at k is the difference between cumulative top-k and cumulative top-$(k-1)$ summed across bid and ask. A uniform grid (each level carries equal depth) puts 0.10 at every level; a fully top-of-book book puts 1.0 at L_1 and zero everywhere above.

The top-of-book carries a median **0.136** of cumulative top-10 depth — a modest premium over the 0.10 uniform null. The remaining 86% is distributed across L_2–L_{10} with a gentle decay from 0.10 at L_2 to $\sim$ 0.08 at L_4 and a roughly flat shoulder at $\sim$ 0.08 through L_{10}. No level captures more than 14% of cumulative depth at the median across markets.

As a single-number shape statistic, the median per-market Kullback–Leibler divergence from the uniform null is **0.087** (IQR $[0.024, 0.474]$, p_{10}–p_{90} $[0.005, 0.985]$). A KL of zero is exactly uniform; a perfectly top-of-book book ($L_1 = 1$, others $= 0$) gives $\log 10 \approx 2.30$. The median panel market sits much closer to uniform than to top-heavy. **9%** of markets are top-heavy in the strong sense ($L_1 > 0.5$); these populate the right tail of L_1-share that overlaid figures often emphasize. **57%** of markets have an L_1 share within $[0.05, 0.20]$, i.e. within a factor-of-two of the uniform null on the most-constrained level.

The folk view that prediction-market books are concentrated at top-of-book does not hold on Polymarket. Depth is layered across all ten levels with a modest top-of-book premium and a long, thin right tail of markets that depart toward top-heavy.

5.3 SF3 – Polygon block-clock alignment

We test whether `price_change` events cluster near Polygon block boundaries by computing, per market, the share of events that fall within ± 100 ms of the nearest 2 000 ms grid point. Under a uniform-timing null this share is 0.10.

The panel-level distribution sits tight against the null: median **0.102** with interquartile

Table 1: SF1 panel — quoted spread by mid-price decile.

Bin	Mid lo	Mid hi	Markets	Median (bps)	p25 (bps)	p75 (bps)
0	0.0000	0.1000	33	1,818	1,176	4,000
1	0.1000	0.2000	18	1,339	689.66	2,564
2	0.2000	0.3000	18	754.99	465.12	1,767
3	0.3000	0.4000	21	2,581	923.08	7,797
4	0.4000	0.5000	184	400.00	202.02	400.00
5	0.5000	0.6000	214	400.00	400.00	400.00
6	0.6000	0.7000	27	444.44	183.49	775.19
7	0.7000	0.8000	12	425.09	139.86	5,915
8	0.8000	0.9000	15	222.22	114.29	591.72
9	0.9000	1.00	4	53.48	0.0000	106.95

Table 2: SF2 — per-level share of cumulative top-10 depth across 547 panel markets with non-null depth. Median and percentile spreads across markets at each book level.

k	Markets	Median share	p25	p75	p10	p90
1	547	0.1364	0.0800	0.2055	0.0243	0.4478
2	547	0.1034	0.0684	0.1364	0.0265	0.1989
3	547	0.0945	0.0641	0.1309	0.0270	0.1820
4	547	0.0806	0.0577	0.0967	0.0217	0.1414
5	547	0.0824	0.0557	0.0966	0.0189	0.1239
6	547	0.0789	0.0353	0.0932	0.0111	0.1165
7	547	0.0818	0.0353	0.1086	0.0100	0.1263
8	547	0.0821	0.0385	0.1039	0.0055	0.1278
9	547	0.0767	0.0258	0.1140	0.0040	0.1383
10	547	0.0830	0.0349	0.1112	0.0028	

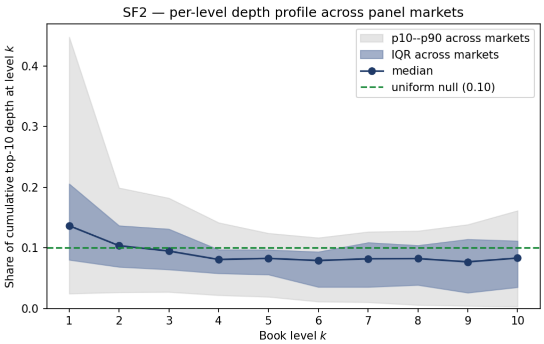

Figure 2: SF2: per-level share of cumulative top-10 depth across 547 panel markets. Solid line is the median; shaded band is the across-market IQR; outer band is p_{10}–p_{90}. The dashed reference line is the uniform null at 0.10 per level.

range $[0.087, 0.110]$; only 27 markets (4.5%) lie above a 0.15 threshold. At the per-market level, however, a two-sided binomial test of H_0: share $= 0.10$ rejects in **351 of 600 markets** (58.5%) at $\alpha = 0.05$. That high rejection rate is a mechanical consequence of large event counts: millions of events per market make the binomial test sensitive to tiny deviations from 0.10. The economic interpretation is that quote-timing does *not* cluster on block boundaries to a meaningful degree, even though high-volume markets reject the exact point null statistically.

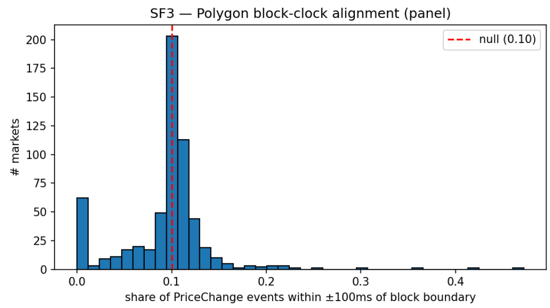

Figure 3: SF3 panel: distribution of per-market block-alignment shares. The red dashed line marks the chance-level null (0.10).

5.4 SF4 – Maker-wallet concentration

For each market we compute the volume-weighted Herfindahl index (HHI) of maker-address share across on-chain trades in the scrape window. A single dominant maker yields HHI $= 1$; a uniform distribution across n

makers yields $1/n$.

Across 600 markets and 6.4 M trades, the median HHI is **0.031** ($\sim$ 32 effective makers). The distribution is right-skewed: $p_{90} = 0.119$ ($\sim$ 8 effective makers) and a maximum of 0.40 (roughly 3 effective makers). Maker liquidity is decentralised on most markets in the panel, with a tail of thin or niche markets dominated by one to three wallets.

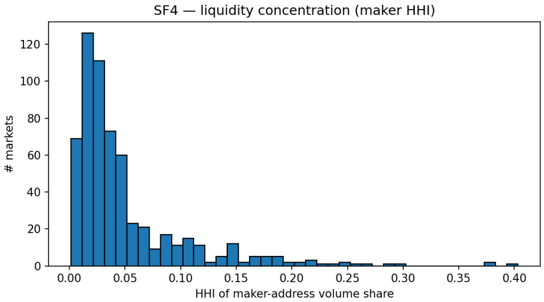

Figure 4: SF4 panel: distribution of per-market maker-address Herfindahl indices across 600 markets.

5.5 SF5 – Category-conditional spread

Per-market metadata is pulled from CLOB REST and questions are keyword-classified into a small bucket scheme. Politics, Business, and Entertainment buckets each had fewer than ten panel markets and were merged into Other for statistical reliability, leaving four categories: Crypto (348/600, 58%), Sports (142/600, 24%), Other (75/600), Geopolitics (35/600). For the on-chain effective half-spread the medians (in probability points) are reported in Table 3 and visualised in Figure 5.

Cross-category dispersion of medians is small in absolute terms (all within ± 0.04 probability points), but within-category interquartile ranges are wide for Crypto and Sports ($p_{75} - p_{25} \approx 0.3$ prob. pp). The wide within-category spread is consistent with the measurement noise documented in Section 7.

5.6 SF6 – Archive-ingestion latency

This stylised fact is a property of our collector pipeline, not of the Polymarket exchange. The two timestamps are emitted on opposite sides of a network and a queue, and their difference

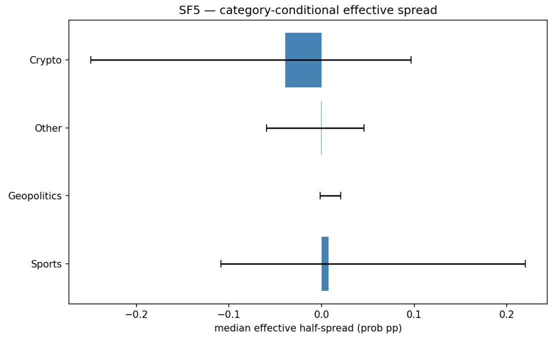

Figure 5: SF5 panel: median effective half-spread by category, with interquartile-range error bars. Categories are derived from keyword classification of CLOB REST `question` text.

reflects whatever the round-trip cost of that pipeline happens to be on a given event. Trader speed enters at most indirectly. Cite SF6 as “archive ingestion latency” rather than “Polymarket latency”.

Each archive row carries two timestamps: `timestamp_received` (exchange side) and `timestamp_created_at` (collector side). Their difference is a per-event ingestion delay. Across 547 markets with non-empty windows, the median per-market p_{50} delay is **41.5 ms**, with p_{90} = 166 ms and p_{99} = 6,108 ms. The inter-market spread of p_{50} is tight (p_{25}–p_{75}: 39–47 ms), which points to collector-pipeline behaviour rather than trader latency. The p_{99} tail of several seconds points to occasional backpressure events at the collector.

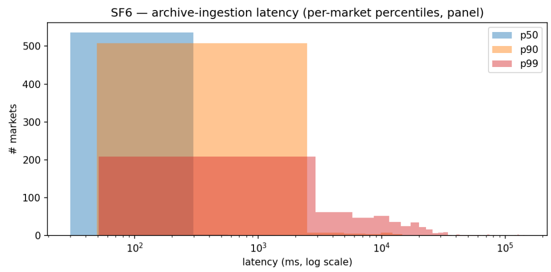

Figure 6: SF6 panel: per-market percentile distributions of archive-ingestion latency (log scale).

5.7 SF7 – Self-counterparty wash share

We flag a trade as wash-suspect under a two-tier rule: (a) `maker == taker`

Table 3: SF5 panel — effective half-spread by keyword-derived category.

Category	Markets	Median half-spread (prob pp)	p25	p75
Sports	142	0.0075	-0.1087	0.2201
Geopolitics	35	0.0001	-0.0014	0.0206
Other	75	-0.0004	-0.0596	0.0456
Crypto	348	-0.0393	-0.2494	0.0968

(direct self-match), or (b) a flipped pair $(\text{maker}_a, \text{taker}_a) \leftrightarrow (\text{taker}_a, \text{maker}_a)$ within 128 blocks (Polygon finality buffer) on the same market. This is an explicit *lower bound*: it captures only direct and immediate-roundtrip self-counterparty patterns, not the extended-graph patterns that network-based classifiers such as Cong et al. [2023] address on unregulated cryptocurrency token exchanges, where wash shares of 25–70% have been documented. We cite this range as a sanity bound, not an apples-to-apples reference: the wash-incentive environment on token exchanges (listing-fee thresholds, aggregator-ranking optimisation, volume-tied market-making rebates) differs from the incentive environment on a prediction market that resolves into a single binary payoff and does not list across competing aggregators.

Across 600 markets and 6.4 M trades, the median wash share is **0.97%**, with $p_{90} = 4.5\%$, $p_{99} = 10.6\%$, and a maximum of 22.2%. The gap between our lower bound and the network-classifier estimates on token exchanges has two components: wash patterns that require multi-counterparty graph analysis to detect (which our detector does not cover), plus a venue-class difference in the underlying wash incentives. We can quantify the first component only under a graph-classifier extension; the second component is identification, not measurement.

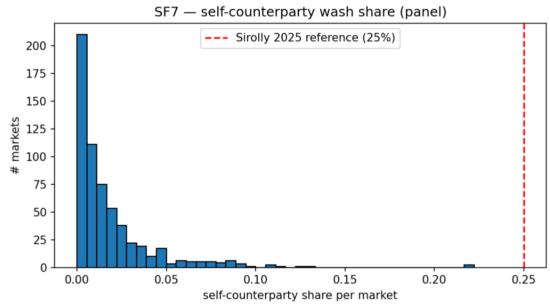

Figure 7: SF7 panel: distribution of self-counterparty wash share by market. Red dashed line marks a 25% reference, the lower bound of the wash-share range documented by Cong et al. [2023] on unregulated cryptocurrency venues.

5.8 SF8 – Depth decay near resolution

Do markets near resolution carry shallower books? We regress log mean depth at $L = 10$ on log seconds-to-close at the panel window midpoint (2026-03-13) under five specifications, restricted to markets with positive seconds-to-close and non-zero summary depth.

Table 4: SF8 — cross-sectional panel regression of log mean depth on log seconds-to-close. HC3-robust standard error. The full v2 spec adds log market duration (window-end minus first-observed event timestamp) and log $p(1-p)$ from the mean window mid; `full_v2_mid_decile_2_7` restricts to mid-price deciles 2–7 (away from the 0/1 boundary).

Spec	n	Slope on log(ttc)	SE (HC3)	R^2
bivariate	322	0.8178	0.1133	0.1293
category_fe	322	0.5500	0.1429	0.2165
category_fe_volume	322	0.3048	0.1036	0.4857
full_v2	313	0.0084	0.1015	0.5702
full_v2_mid_decile_2_7	187	-0.2702	0.1687	0.2909

The v1 trio (rows 1–3 of Table 4) shows

a positive slope that attenuates from **0.818** (bivariate) to **0.550** with category fixed effects, then to **0.305** with log volume added on top. A reader could conclude from these three rows that markets nearer resolution are shallower in the cross-section. They would be wrong: the v1 trio omits two confounds that turn out to absorb the time-to-close coefficient entirely.

The full v2 specification adds log market duration (window-end minus the first-observed-event timestamp, left-censored at the archive start) and log $p(1-p)$, computed from each market's mean mid-price across the window. With those alongside log volume and category fixed effects, the log seconds-to-close coefficient collapses to +0.008 (HC3 SE 0.102, $t = 0.08$, $R^2 = 0.57$, $n = 313$). Restricting the sample to mid-price deciles 2–7, which removes markets whose price is close enough to 0 or 1 that $p(1-p)$ is itself small, the coefficient is -0.270 (HC3 SE 0.169, $t = -1.60$, $n = 187$) — again indistinguishable from zero.

What the full v2 spec identifies are three structural drivers of cross-sectional depth: log duration enters at +0.222 (SE 0.073; longer-lived markets accumulate more depth); log $p(1-p)$ enters at -1.02 (SE 0.180; markets at price extremes carry more depth on average, consistent with mature markets that have settled toward resolution); log volume enters at +0.41 (SE 0.046). Time-to-close itself adds no explanatory power once those three are conditioned on. The v1 trio's positive slope was mechanical mediation: time-to-close is correlated with duration (longer-lived markets sit further from resolution in the cross-section by construction) and with price state (volatile new markets cluster near $p = 0.5$).

The honest cross-sectional reading: depth on Polymarket is explained by duration, price level, and volume, not by proximity to resolution per se. A within-market trajectory test — does an individual market's depth fall as it approaches its own resolution? — is identified only by per-market depth time-series, which the panel does not materialise. That question is deferred.

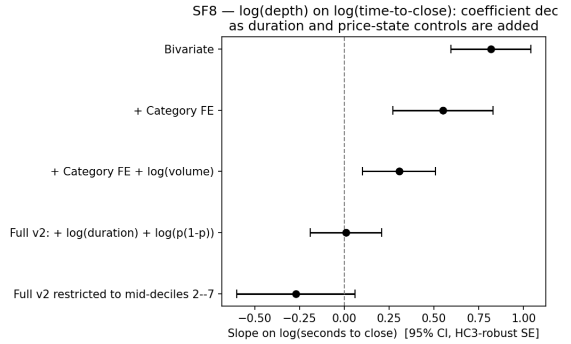

Figure 8: SF8: coefficient on log(seconds-to-close) across five specifications, with 95% HC3-robust confidence intervals. The v1 trio (top three) shows a significantly positive slope; the full v2 spec (fourth) collapses it to zero once duration and $p(1-p)$ are conditioned on.

6 Spread Decomposition

Following Glosten and Harris [1988] and the modern restatements in Huang and Stoll [1997], Madhavan et al. [1997], we decompose the per-market effective half-spread into two components:

$$S_{1/2}^{\text{eff}} = c + \varphi, \tag{1}$$

where c is a transitory order-processing / inventory component, recovered as the realized half-spread at a 60-second lag, and φ is the residual adverse-selection component.

We restrict the decomposition to the top-100 stratum, where on-chain trade counts are highest (median 11,000 trades per market in window). Per-market values come from `data/panel_trade_measures.parquet`, computed by injecting authoritative on-chain trades into the measure pipeline described in Section 7. The decomposition converges on 97 of 100 markets; the three non-converging markets had too few signed trades after on-chain alignment for the realized-half-spread term to identify.

The median effective half-spread on the top-100 panel is essentially zero (-0.0003 prob pp), as are the median transitory component (0.00001) and the median adverse-selection component (0.0). This near-null pattern lines up with the calibration in Section 7: once sign errors are removed, the dollar-weighted "adverse selection" that orderbook-only inference produces collapses, leaving the

Table 5: Glosten-Harris decomposition — first 10 top-100 markets (hex-ordered) with a converged decomposition, in probability points. 97 of 100 panel markets converged; three did not.

Market id (truncated)	Effective half	c (transitory)	φ (adverse sel.)
0x03bc660a4df5fa...	-0.2431	-0.2431	-0.0000
0x06707a5317654a...	0.0778	0.0778	0.0000
0x07d45de444dbe0...	0.0267	0.0270	-0.0004
0x092471f61558da...	-0.6657	-0.6657	-0.0000
0x0954cc08de0ab8...	-0.1087	-0.1087	0.0000
0x09cbe3e796661a...	-0.0205	-0.0128	-0.0077
0x09e8f0db05570a...	0.0046	0.0052	-0.0006
0x0b4cc3b739e1df...	0.0002	0.0002	0.0000
0x119fa68324e678...	0.2201	0.2201	0.0000
0x1745319c9ef1e4...	-0.3392	-0.3392	0.0000

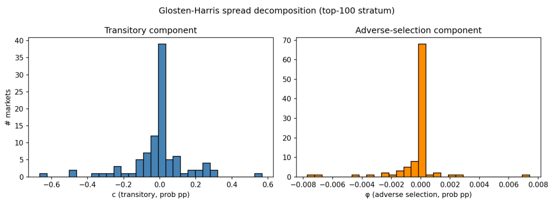

Figure 9: Glosten-Harris decomposition across the top-100 stratum: distribution of transitory component c (left) and adverse-selection component φ (right), both in probability points.

typical top-100 market with no detectable systematic spread component on either side. The distribution tails carry both market-specific adverse-selection events and residual measurement noise from the 60-second sample-step compromise documented in Section 7.4.

7 Limits of Orderbook-Only Inference

7.1 Sign-agreement against on-chain ground truth

Six trade-based measures (effective spread, realized spread, Roll, Abdi-Ranaldo, Kyle's λ, Amihud) require an aggressor sign for each trade. Standard practice in equity microstructure infers that sign from a quote-driven feed via Lee-Ready, which assumes the feed exposes enough information to distinguish buyer-initiated from seller-initiated trades.

The order-book event stream from Polymarket's public WebSocket — `price_change` and `book_snapshot` updates — does not, by itself. We infer trades from this stream under a LOOSE rule (every resting-size decrement counts) and match the inferred buckets against on-chain `OrderFilled` events at 5 s and exact-price granularity. We run this matching over four disjoint 7-day windows (2026-02-28 – 03-06, 03-07 – 03-13, 03-14 – 03-20, 03-21 – 03-27) on the top-100 stratum, computing sign-agreement and a 95% bootstrap CI for cells with at least ten matched buckets.

- Panel mean (109 valid cells of 400, 55 markets): **0.615**, market-clustered bootstrap 95% CI $[0.579, 0.653]$
- Volume-weighted by matched-bucket count (total 125,080): **0.592**, market-clustered bootstrap 95% CI $[0.542, 0.659]$
- Panel median 0.591; IQR $[0.526, 0.681]$; p_{10}–p_{90} $[0.464, 0.841]$
- Per-window medians 0.586 / 0.591 / 0.618 / 0.645

A volume-weighted sign-agreement of $\sim$ 59% sits just above the 50% chance baseline. Even when an inferred bucket matches an on-chain bucket in time and price, the inferred aggressor direction is wrong about two trades in five. The mechanism is the order-book event stream itself: `price_change` updates broadcast a post-match snapshot of the resting book without identifying the taker. The `change_side` field marks which side of the book moved, not which side initiated the trade, and using it as a sign proxy produces the $\sim$ 59% agreement rate. Polymarket's market channel does additionally emit a `last_trade_price` event; evaluating its agreement with on-chain ground truth is an obvious follow-up, and is out of scope for the present archive.

7.2 Propagation to direction-dependent measures

A noisy sign propagates to every measure that consumes it. We compute all six trade-based measures under STRICT inference and under authoritative on-chain trades on the top-100 stratum over a 7-day window (2026-03-07 – 03-14); the window is shorter than the full scrape because STRICT inference is $O(n^2)$ in the recent-event list and the 28-day version did not finish in tractable wall time on this panel size.

Three patterns emerge from the panel distribution.

Trade-count divergence is direction-unpredictable. Among 31 markets where both inference paths yielded non-empty trade streams, the STRICT/on-chain count ratio has a median of 0.23 (p_{10}–p_{90} $[0.01, 1.45]$; min–max $[0.004, 3.62]$). STRICT typically under-counts by an order of magnitude; on a small tail of markets it over-counts by up to 3.6×. There is no correctable scaling factor.

Effective spread sign-flips on two-thirds of the comparable panel. Among 24 markets with both sources non-empty, 16 (67%, Wilson 95% CI $[0.47, 0.82]$) have STRICT and on-chain effective half-spreads of opposite sign. The STRICT median is +0.0048 prob pp; the on-chain median is −0.000075 prob pp. Inferred trade directions induce a positive spread that the authoritative trade record does not support.

Kyle's λ sign-flips on three-fifths of the comparable panel. Of 25 markets with both sources non-null and non-NaN, 15 (60%, Wilson 95% CI $[0.41, 0.77]$) have STRICT and on-chain Kyle's λ of opposite sign.

Selection robustness. Only 24–31 of the 100 top markets had both inference paths non-empty in the 7-day slice (the remainder had zero events on one side); the comparable-market subset is what supports the sign-flip rates above. Three checks address whether the subset is representative. First, the 24 effective-spread markets cover **98% of top-100 on-chain dollar volume** on this window, and the 25 Kyle's λ markets cover **99%** — the dropped markets are the small, quiet tail rather than typical activity. Second, weighting by on-chain dollar volume across the comparable subset, the effective-spread sign-flip rate **rises** from 67% to **86%** (bootstrap 95% CI $[0.32, 0.99]$): the largest market in the window, which alone accounts for two-thirds of top-100 volume, is one of the markets that sign-flips. Volume weighting therefore strengthens the headline rather than softening it. Kyle's λ moves the other way: weighting by on-chain dollar volume drops the sign-flip share from 60% to **18%** (bootstrap 95% CI $[0.04, 0.81]$), because the λ estimate on the largest market is dominated by zero-bucket noise where both sources happen to agree on a near-zero negative value. The Kyle's λ instability is concentrated on smaller markets where λ is fragile in any case (Section 7.4); the effective-spread result is the more load-bearing one. Restricting the effective-spread comparison to markets with at least 100 STRICT trades pushes the unweighted sign-flip rate from 67% to 73% (16/22, Wilson 95% CI $[0.52, 0.87]$); the Kyle's λ rate is essentially unchanged (59% at the 100-bucket threshold). Third, we re-run the comparison on the next non-overlapping 7-day window (2026-03-14 to 03-21). Comparable subsets are larger in this window (30 markets for both effective spread and Kyle's λ). The unweighted sign-flip rates fall by about 17 percentage points: the effective-spread rate is 50% (15/30, Wilson 95% CI $[0.33, 0.67]$) and the Kyle's λ rate is 43% (13/30, Wilson 95% CI $[0.27, 0.61]$). Both Wilson intervals contain the 50% chance baseline. The two-window swing is meaningful: STRICT-vs-on-chain sign agreement on these measures is window-specific. What does not swing is the methodological conclusion. Across both windows the flip rate sits in or below the chance band, the published headline (67% / 60% on window A) is not exceptional, and on neither window does feed-based inference recover the on-chain sign at the $\sim$ 80% rate that Lee-Ready achieves on equity venues. Window-B results ship in `measures_compare_top100_window_b.parquet` and the side-by-side in `r3_window_robustness.csv`.

7.3 Implications for Polymarket microstructure research

Polymarket microstructure results that depend on trade direction require on-chain `OrderFilled` events as the direction source, not the public order-book feed alone. We release a replication package (`polydata.onchain.trades.load_onchain_trades`) that performs the off-chain to on-chain join, together with a small patch set (`polydata.measures._trade_source.resolve_trades`) that lets existing measure code accept injected authoritative trades without a rewrite. The same constraint should bind on any decentralised CLOB venue where the off-chain matching layer broadcasts post-match book state without exposing the taker identity (GMX v1, dYdX v3, Loopring's historical CLOB, and similar hybrid architectures): the public order-book feed exposes *what cleared* but not *who initiated*, and any direction-dependent microstructure measure on those venues will need an authoritative on-chain trade source. The replication package is contract-agnostic on the trade-loading side; only the venue-specific event signature needs to be plugged in.

7.4 Methodology caveats

Three places where our compute approach trades accuracy for tractability on a 28-day, 600-market panel.

Sample step. Quote samples and bucketed lookups use a 60-second grid for the panel compute. Sensitivity tests on the top-5 markets across $\{1, 10, 60, 300\}$ s (artifact `sample_step_sensitivity.parquet`) show that Roll is invariant by construction, that effective spread and Amihud are stable to within $\sim$ 10–20% at 60 s relative to 1 s on most markets, and that Kyle's λ varies by orders of magnitude across step sizes on noisy markets. Kyle's λ is the most fragile estimator at any sample step, consistent with its sign-flip behaviour in Section 7.2.

Cross-sectional SF8. SF8 (Section 5.8) is cross-sectional at a single panel midpoint. We cannot, from one snapshot, separate within-market trajectory effects (depth falling as a given market approaches its own resolution) from between-market structural differences (longer-lived, higher-volume markets sit deeper on average). The full v2 specification conditions on the between-market factors that are identified at one snapshot — duration, price level, volume, category — and the residual time-to-close coefficient is statistically indistinguishable from zero. A proper within-market temporal regression would require per-market depth time-series that the current panel does not materialise. We therefore report SF8 as a structural cross-sectional finding about duration, price state, and volume, not as a time-to-resolution effect.

Wash-filter lower bound. SF7 (Section 5.7) returns a lower bound by construction: the detector covers direct self-match and one-step roundtrip patterns, not the multi-counterparty graph patterns that network-based classifiers detect. Closing the gap on Polymarket specifically would require a graph-classifier extension; the Cong et al. [2023] 25–70% range is a token-exchange sanity bound, not an apples-to-apples benchmark for prediction markets.

MEV as a confound on the buyer/seller asymmetry. Polymarket settles on Polygon, whose mempool is publicly accessible to bots that can sandwich incoming orders, place runner trades, or execute related back-running strategies. We do not analyse mempool data, so the on-chain trade record we treat as ground truth contains whatever fills MEV bots produce on top of organic flow. The 60/40 seller/buyer asymmetry across the panel is therefore consistent with retail aggressor behaviour, MEV bot activity, or some mixture of the two. Decomposing it would help via Polygon mempool capture during the scrape window or a wallet-clustering approach against known MEV searchers; both are outside the scope of this paper. Public-mempool capture on Polygon is not a clean solution either, because a non-trivial share of order flow on Polygon routes through private mempools (Marlin Relay, Bloxroute, Flashbots-on-Polygon variants) and never appears in the public feed to begin with. A future MEV-attribution exercise would need to combine public-mempool data with relay-side captures from at least the largest private channels.

8 Conclusion

This paper joins a tick-level Polymarket order-book archive to the authoritative on-chain trade record and reports cross-sectional microstructure for a pre-registered 600-market panel. On the quoted side we find a longshot spread premium, a depth profile closer to a uniform geometric grid than the top-of-book pattern usually assumed for prediction markets, and an archive-ingestion latency distribution whose tight inter-market spread points to collector behaviour rather than trader speed. On the on-chain side, maker liquidity is broadly decentralised with a thin tail dominated by a few wallets, and self-counterparty wash activity is low under our direct-detection lower bound.

The methodological finding is the more consequential one. Trade direction inferred from Polymarket's public WebSocket feed agrees with on-chain ground truth on only $\sim 59\%$ of comparable buckets, barely above the chance baseline. On the comparable subset of the top-100 panel the propagation is window-specific but consistent in direction: the effective half-spread flips sign on 67% of markets in our first 7-day window and 50% in a second non-overlapping window, with Kyle's λ at 60% and 43% respectively. Both windows sit at or below the chance band, well short of the $\sim 80\%$ Lee-Ready accuracy on equity venues. Polymarket microstructure work that depends on aggressor sign therefore needs to source it from `OrderFilled` events rather than the public feed.

Four extensions follow naturally from the data we did not exploit. The CTF Exchange V1 to V2 cutover at the end of April 2026 closes our scrape window and opens a venue-evolution comparison. A combined public-mempool plus private-relay capture would help separate MEV from retail aggressor flow in the 60/40 seller/buyer asymmetry, with the caveat that a non-trivial share of Polygon order flow routes through private channels that public-mempool capture alone misses. Per-market depth time series, which our panel summarises away, would let SF8 move from a cross-sectional to a within-market depth-decay regression. And cross-venue analysis against Kalshi, PredictIt, or sports-book mirrors would address the price-discovery question we leave open here: are Polymarket's prices the leader or the follower for the real-world events its markets resolve on?

9 Disclosures

9.1 Use of AI

The data-collection infrastructure and analysis code were developed with assistance from AI-based coding tools (Claude Code CLI with Claude Opus 4.7). The statistical analysis, interpretation of results, and all written content in the manuscript represent the original intellectual contributions of the author. AI writing assistants were used for grammar and clarity improvements.

9.2 Conflicts of Interest

The author declares no conflicts of interest.

9.3 Ethics Statement

This study analyzes publicly observable orderbook and trade data. The Polymarket public WebSocket feed broadcasts limit-order-book state updates that any client can subscribe to; the Polygon `OrderFilled` events are public on-chain transactions recorded by the Polymarket CTF Exchange smart contract. No private user data was collected. Counterparty addresses on chain are pseudonymous public-key identifiers; we use them only to compute maker-wallet concentration (Section 5.4) and a self-counterparty wash-share lower bound (Section 5.7), without attempting to deanonymise any wallet.

9.4 Data and Code Availability

The complete analysis code and per-market panel artifacts are available at `https://github.com/philippdubach/polymarket-microstructure` and archived at Zenodo under DOI 10.5281/zenodo.19811426. The 30-billion-event raw orderbook archive (623.8 GB) is not redistributed in the replication package due to size; access can be arranged via the corresponding author. The on-chain `OrderFilled` scrape is reproducible from the `scripts/scrape_onchain_fills.py` pipeline against any Polygon RPC provider with archive-node access.

References

Jonathan Brogaard, Terrence Hendershott, and Ryan Riordan. High-frequency trading and price discovery. *Review of Financial Studies*, 27(8):2267–2306, 2014.

Lin William Cong, Xi Li, Ke Tang, and Yang Yang. Crypto wash trading. *Management Science*, 69(11):6427–6454, 2023.

Philipp D. Dubach. Replication package: The anatomy of a decentralized prediction market, 2026. URL `https://doi.org/10.5281/zenodo.19811426`.

Katrina Ellis, Roni Michaely, and Maureen O'Hara. The accuracy of trade classification rules: Evidence from nasdaq. *Journal of Financial and Quantitative Analysis*, 35(4): 529–551, 2000.

Thierry Foucault, Marco Pagano, and Ailsa Röell. *Market Liquidity: Theory, Evidence, and Policy*. Oxford University Press, 2013.

Lawrence R. Glosten and Lawrence E. Harris. Estimating the components of the bid/ask spread. *Journal of Financial Economics*, 21 (1):123–142, 1988.

Robin Hanson. Logarithmic market scoring rules for modular combinatorial information aggregation. *Journal of Prediction Markets*, 1(1):3–15, 2007.

Joel Hasbrouck. *Empirical Market Microstructure: The Institutions, Economics, and Econometrics of Securities Trading*. Oxford University Press, 2007.

Roger D. Huang and Hans R. Stoll. The components of the bid-ask spread: A general approach. *Review of Financial Studies*, 10 (4):995–1034, 1997.

Charles M. C. Lee and Mark J. Ready. Inferring trade direction from intraday data. *Journal of Finance*, 46(2):733–746, 1991.

Ananth Madhavan, Matthew Richardson, and Mark Roomans. Why do security prices change? a transaction-level analysis of NYSE stocks. *Review of Financial Studies*, 10(4):1035–1064, 1997.

Charles F. Manski. Interpreting the predictions of prediction markets. *Economics Letters*, 91 (3):425–429, 2006.

Maureen O'Hara. *Market Microstructure Theory*. Blackwell Publishing, 1995.

Lionel Page and Robert T. Clemen. Do prediction markets produce well-calibrated probability forecasts? *The Economic Journal*, 123(568):491–513, 2013.

Nahid Rahman, Joseph Al-Chami, and Jeremy Clark. SoK: Market microstructure for decentralized prediction markets (DePMs). *arXiv preprint arXiv:2510.15612*, 2025. URL `https://arxiv.org/abs/2510.15612`.

Erik Snowberg and Justin Wolfers. Explaining the favorite-long shot bias: Is it risk-love or misperceptions? *Journal of Political Economy*, 118(4):723–746, 2010.

Richard H. Thaler and William T. Ziemba. Anomalies: Parimutuel betting markets: Racetracks and lotteries. *Journal of Economic Perspectives*, 2(2):161–174, 1988.

Kwok Ping Tsang and Zichao Yang. The anatomy of Polymarket: Evidence from the 2024 presidential election. *arXiv preprint arXiv:2603.03136*, 2026. URL `https://arxiv.org/abs/2603.03136`.

Justin Wolfers and Eric Zitzewitz. Prediction markets. *Journal of Economic Perspectives*, 18(2):107–126, 2004.